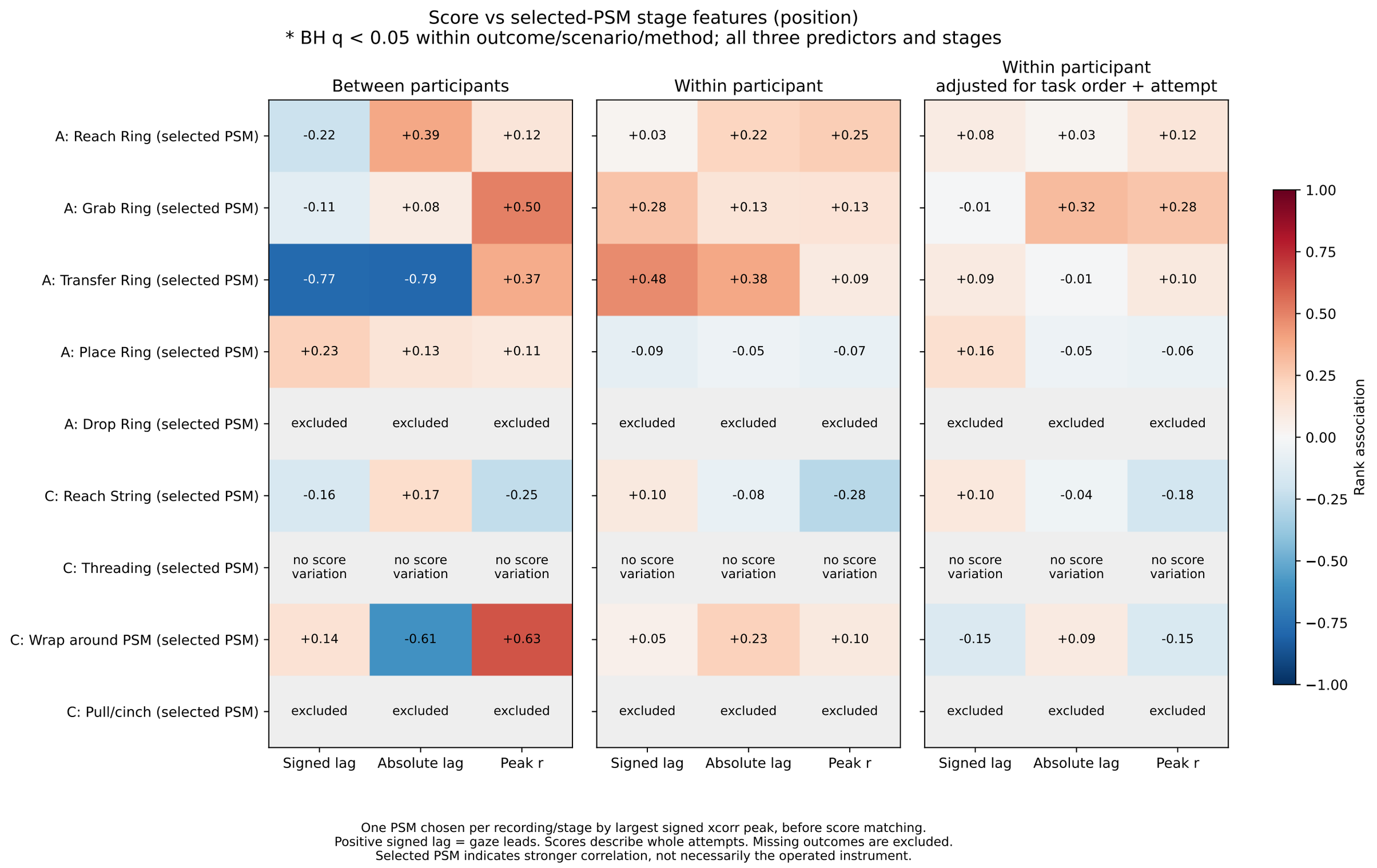
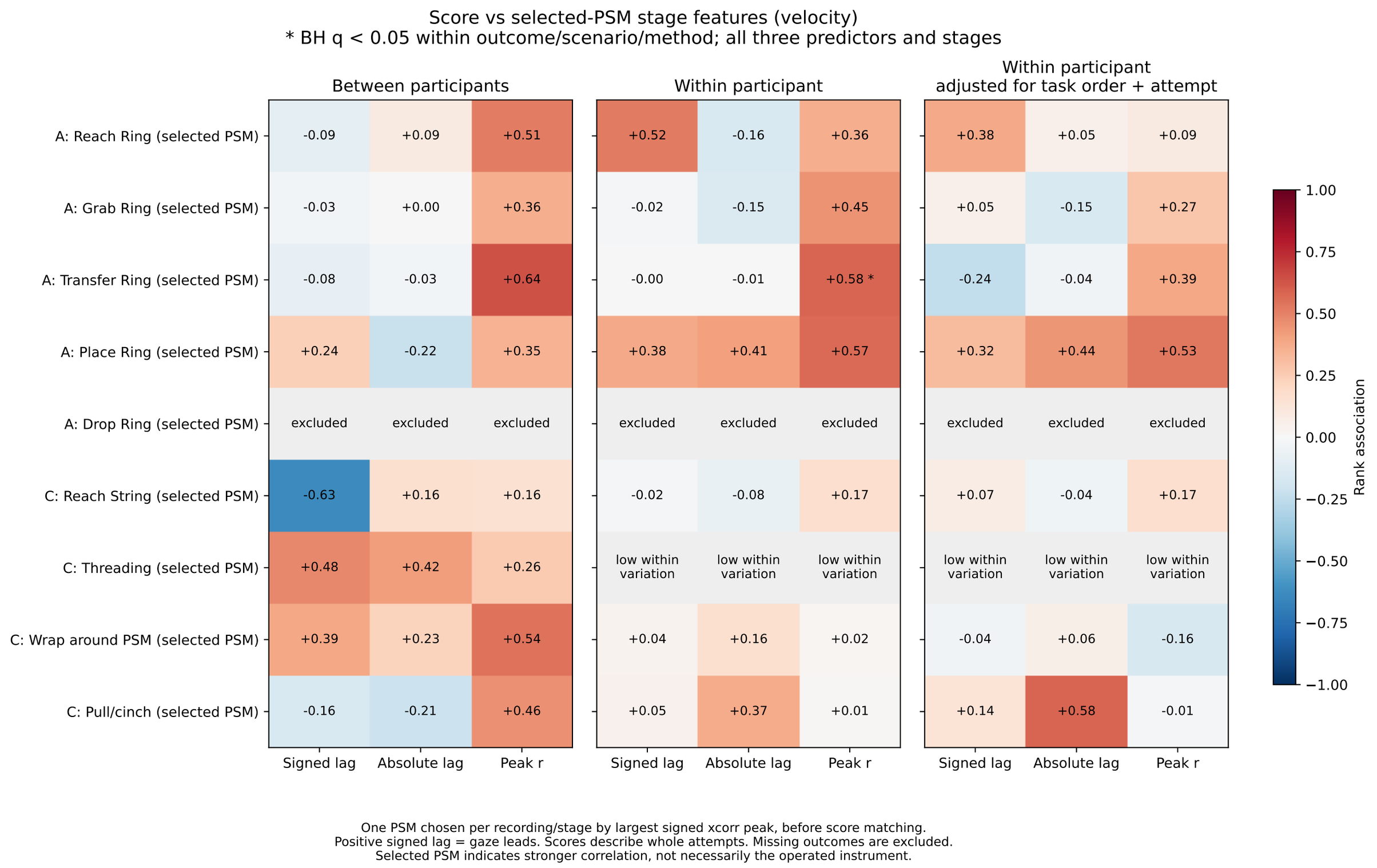
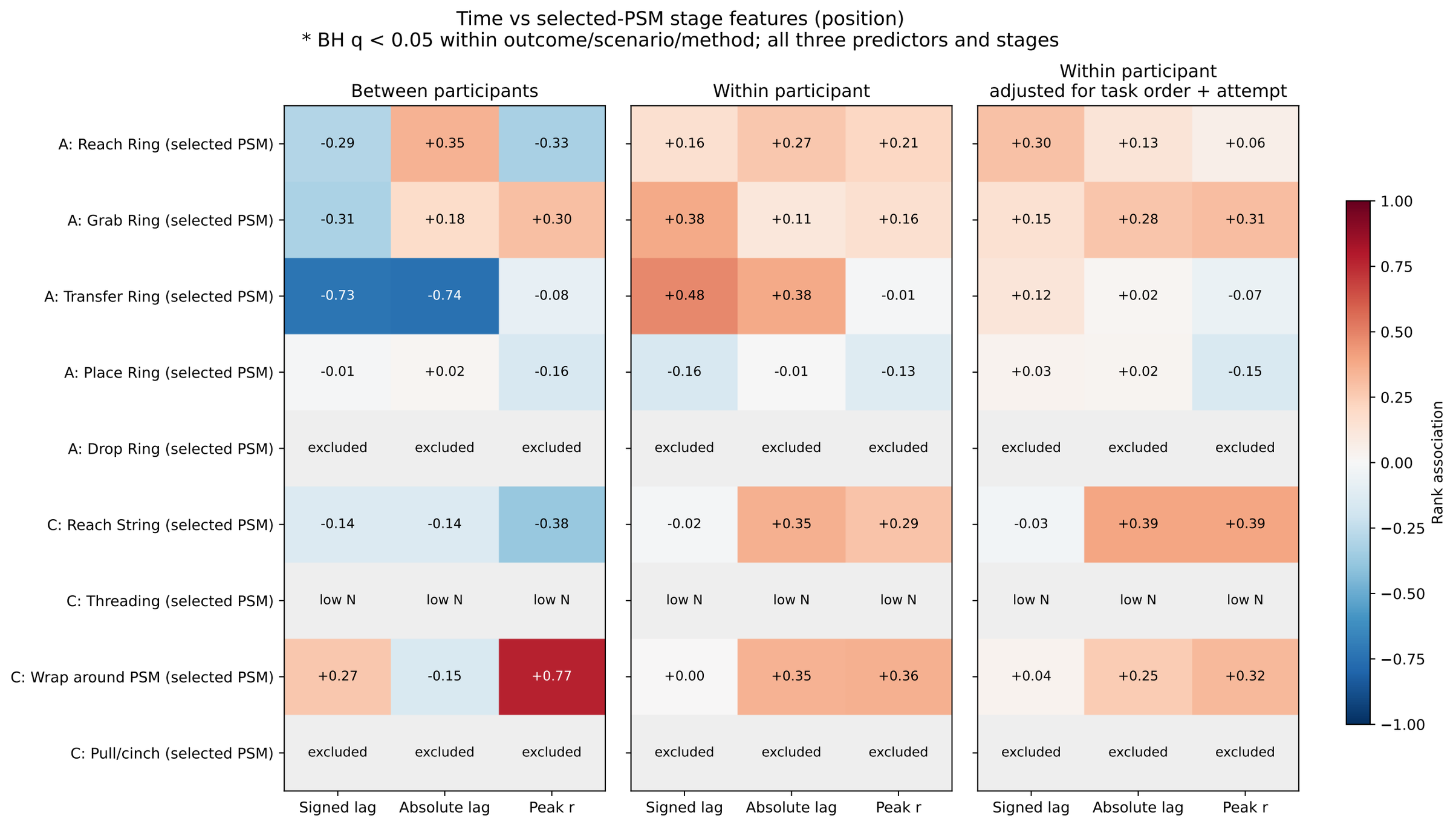
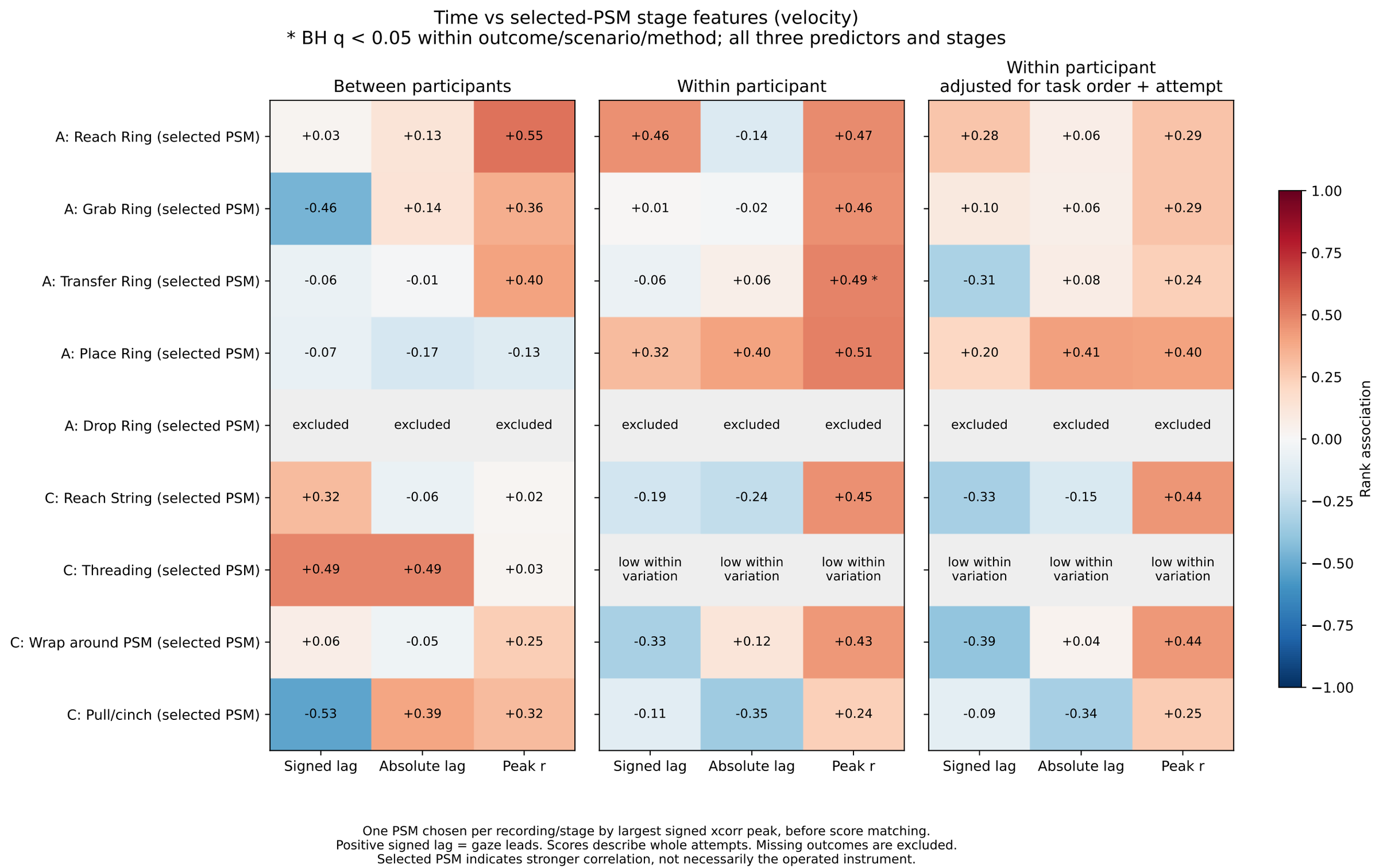

Score vs selected-PSM stage features (position)
* BH q < 0.05 within outcome/scenario/method; all three predictors and stages
Between participants
Within participant
Within participant
adjusted for task order + attempt
A: Reach Ring (selected PSM)
A: Grab Ring (selected PSM)
A: Transfer Ring (selected PSM)
A: Place Ring (selected PSM)
A: Drop Ring (selected PSM)
C: Reach String (selected PSM)
C: Threading (selected PSM)
C: Wrap around PSM (selected PSM)
C: Pull/cinch (selected PSM)
-0.22 +0.39 +0.12
+0.03 +0.22 +0.25
+0.08 +0.03 +0.12
-0.11 +0.08 +0.50
+0.28 +0.13 +0.13
-0.01 +0.32 +0.28
-0.77 -0.79 +0.37
+0.48 +0.38 +0.09
+0.09 -0.01 +0.10
+0.23 +0.13 +0.11
-0.09 -0.05 -0.07
+0.16 -0.05 -0.06
excluded excluded excluded
excluded excluded excluded
excluded excluded excluded
-0.16 +0.17 -0.25
+0.10 -0.08 -0.28
+0.10 -0.04 -0.18
no score variation
+0.14 -0.61 +0.63
+0.05 +0.23 +0.10
-0.15 +0.09 -0.15
excluded excluded excluded
excluded excluded excluded
excluded excluded excluded
Signed lag
Absolute lag
Peak r
Rank association
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
One PSM chosen per recording/stage by largest signed xcorr peak, before score matching.
Positive signed lag = gaze leads. Scores describe whole attempts. Missing outcomes are excluded.
Selected PSM indicates stronger correlation, not necessarily the operated instrument.

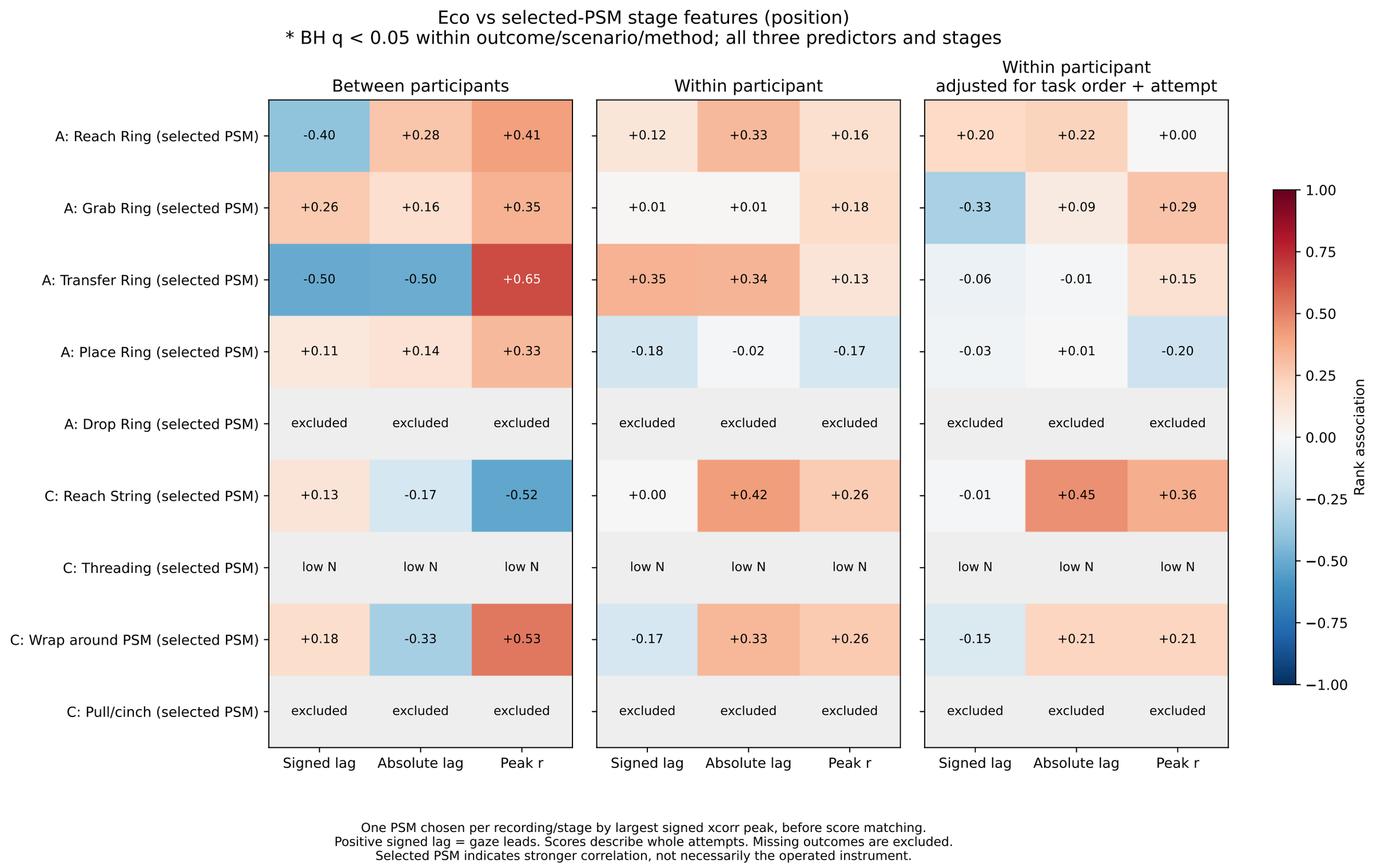


One PSM chosen per recording/stage by largest signed xcorr peak, before score matching.
Positive signed lag = gaze leads. Scores describe whole attempts. Missing outcomes are excluded.
Selected PSM indicates stronger correlation, not necessarily the operated instrument.

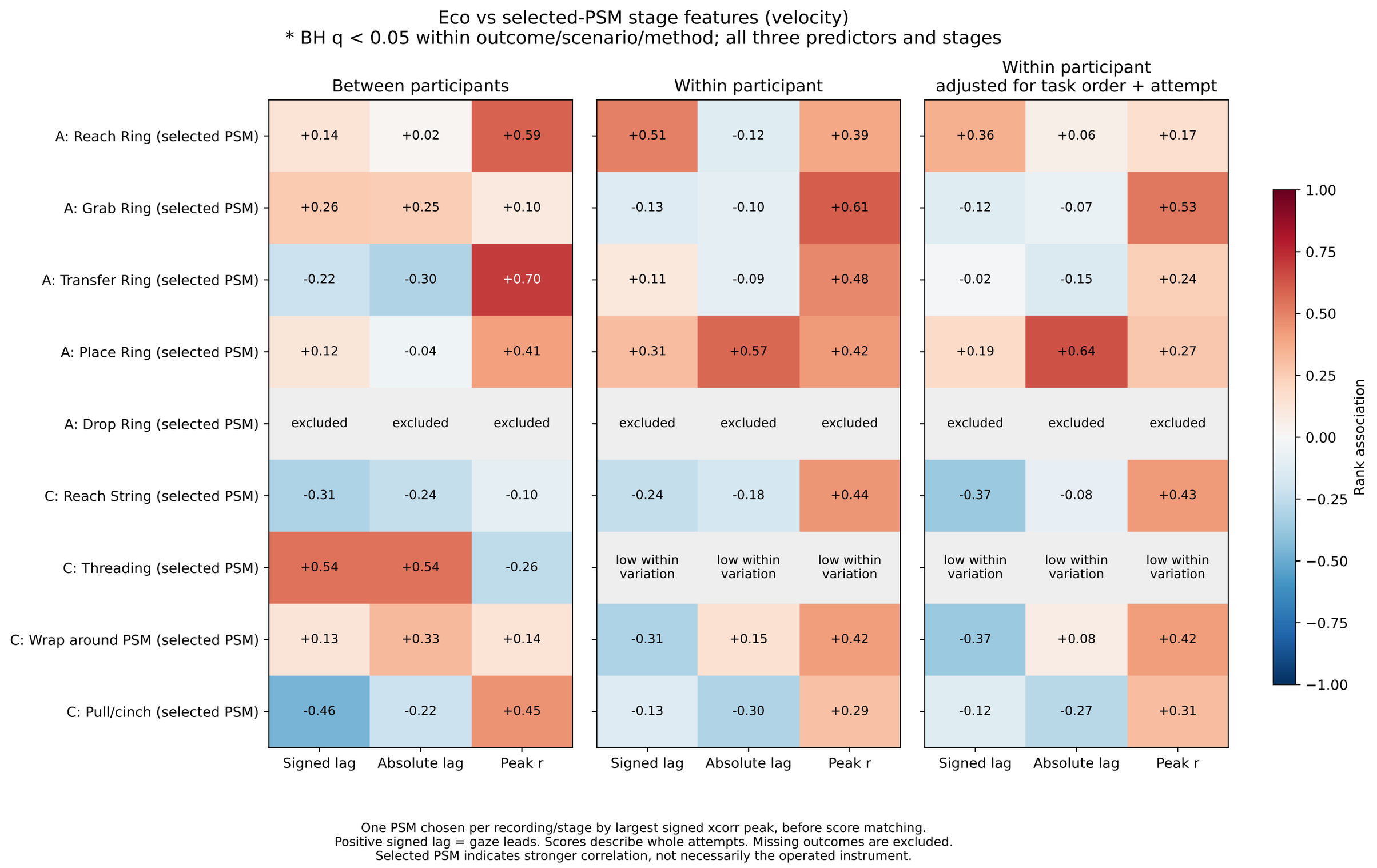


One PSM chosen per recording/stage by largest signed xcorr peak, before score matching.
Positive signed lag = gaze leads. Scores describe whole attempts. Missing outcomes are excluded.
Selected PSM indicates stronger correlation, not necessarily the operated instrument.

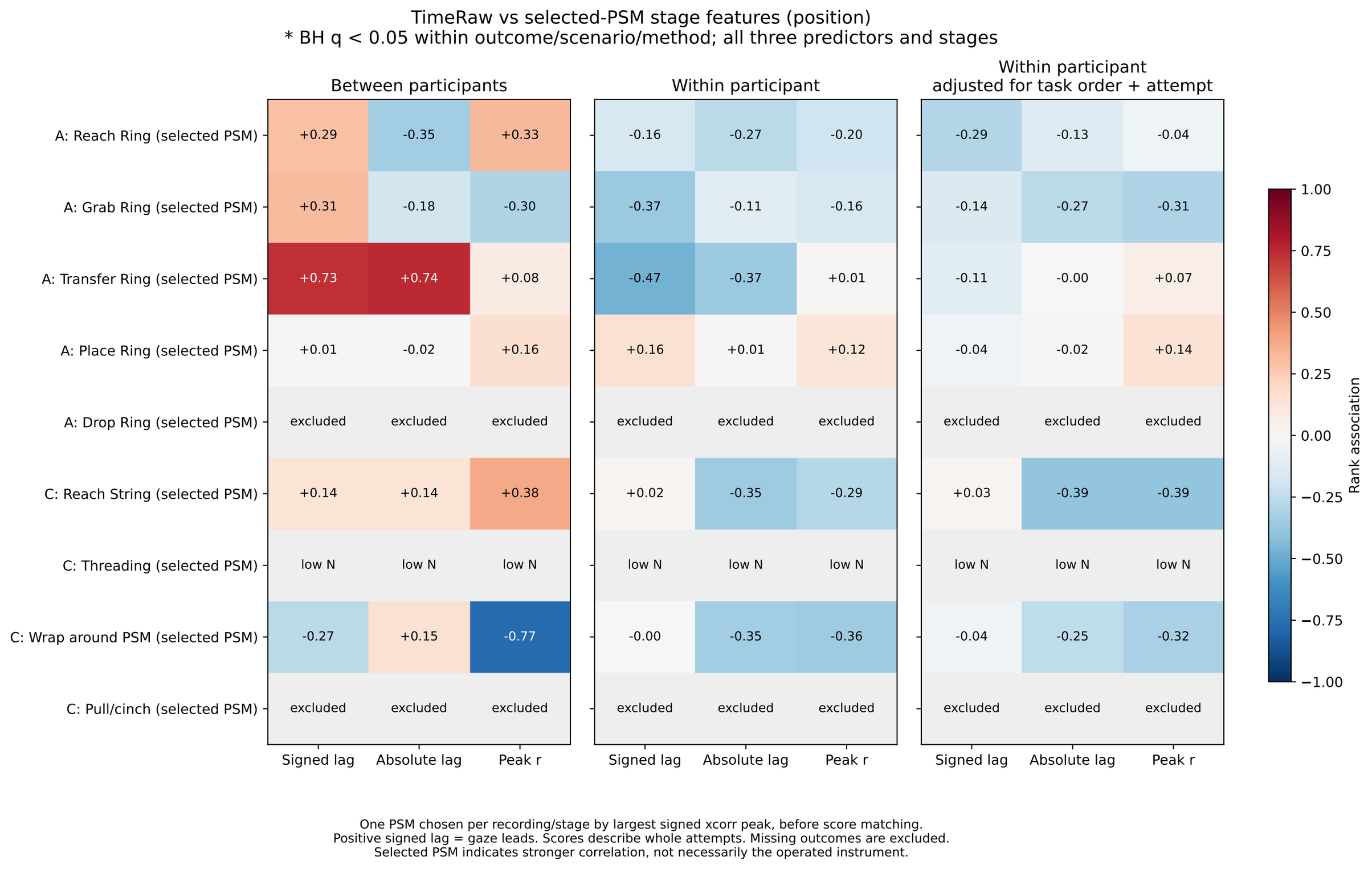

Time vs selected-PSM stage features (velocity)
* BH q < 0.05 within outcome/scenario/method; all three predictors and stages
Between participants
Within participant
Within participant
adjusted for task order + attempt
A: Reach Ring (selected PSM)
A: Grab Ring (selected PSM)
A: Transfer Ring (selected PSM)
A: Place Ring (selected PSM)
A: Drop Ring (selected PSM)
C: Reach String (selected PSM)
C: Threading (selected PSM)
C: Wrap around PSM (selected PSM)
C: Pull/cinch (selected PSM)
+0.03 +0.13 +0.55
-0.46 +0.14 +0.36
-0.06 -0.01 +0.40
-0.07 -0.17 -0.13
excluded excluded excluded
+0.32 -0.06 +0.02
+0.49 +0.49 +0.03
+0.06 -0.05 +0.25
-0.53 +0.39 +0.32
+0.46 -0.14 +0.47
+0.01 -0.02 +0.46
-0.06 +0.06 +0.49 *
+0.32 +0.40 +0.51
excluded excluded excluded
-0.19 -0.24 +0.45
low within variation
low within variation
low within variation
-0.33 +0.12 +0.43
-0.11 -0.35 +0.24
+0.28 +0.06 +0.29
+0.10 +0.06 +0.29
-0.31 +0.08 +0.24
+0.20 +0.41 +0.40
excluded excluded excluded
-0.33 -0.15 +0.44
low within variation
low within variation
low within variation
-0.39 +0.04 +0.44
-0.09 -0.34 +0.25
Signed lag
Absolute lag
Peak r
Signed lag
Absolute lag
Peak r
Signed lag
Absolute lag
Peak r
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Rank association
One PSM chosen per recording/stage by largest signed xcorr peak, before score matching.
Positive signed lag = gaze leads. Scores describe whole attempts. Missing outcomes are excluded.
Selected PSM indicates stronger correlation, not necessarily the operated instrument.

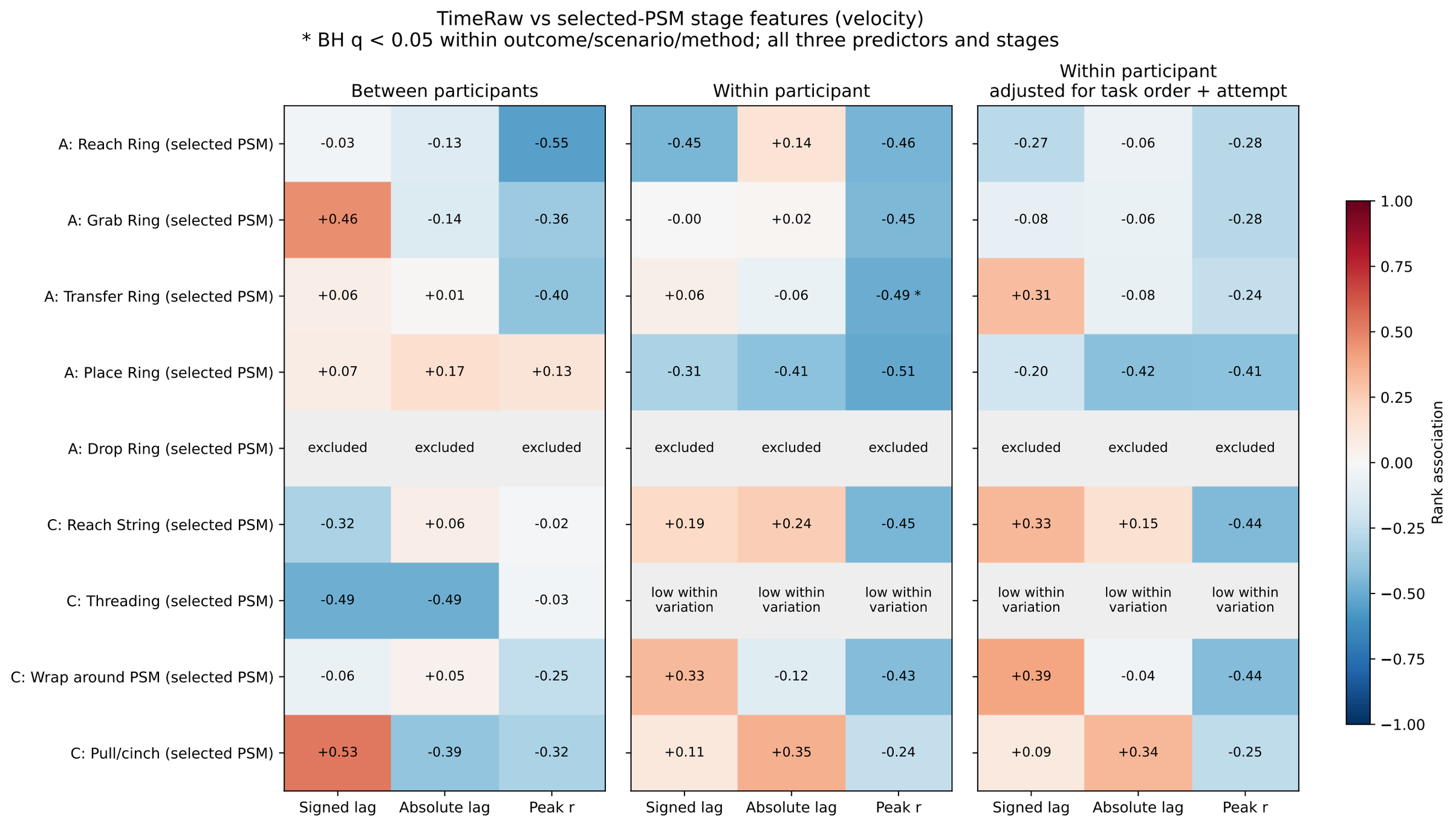


One PSM chosen per recording/stage by largest signed xcorr peak, before score matching.
Positive signed lag = gaze leads. Scores describe whole attempts. Missing outcomes are excluded.
Selected PSM indicates stronger correlation, not necessarily the operated instrument.

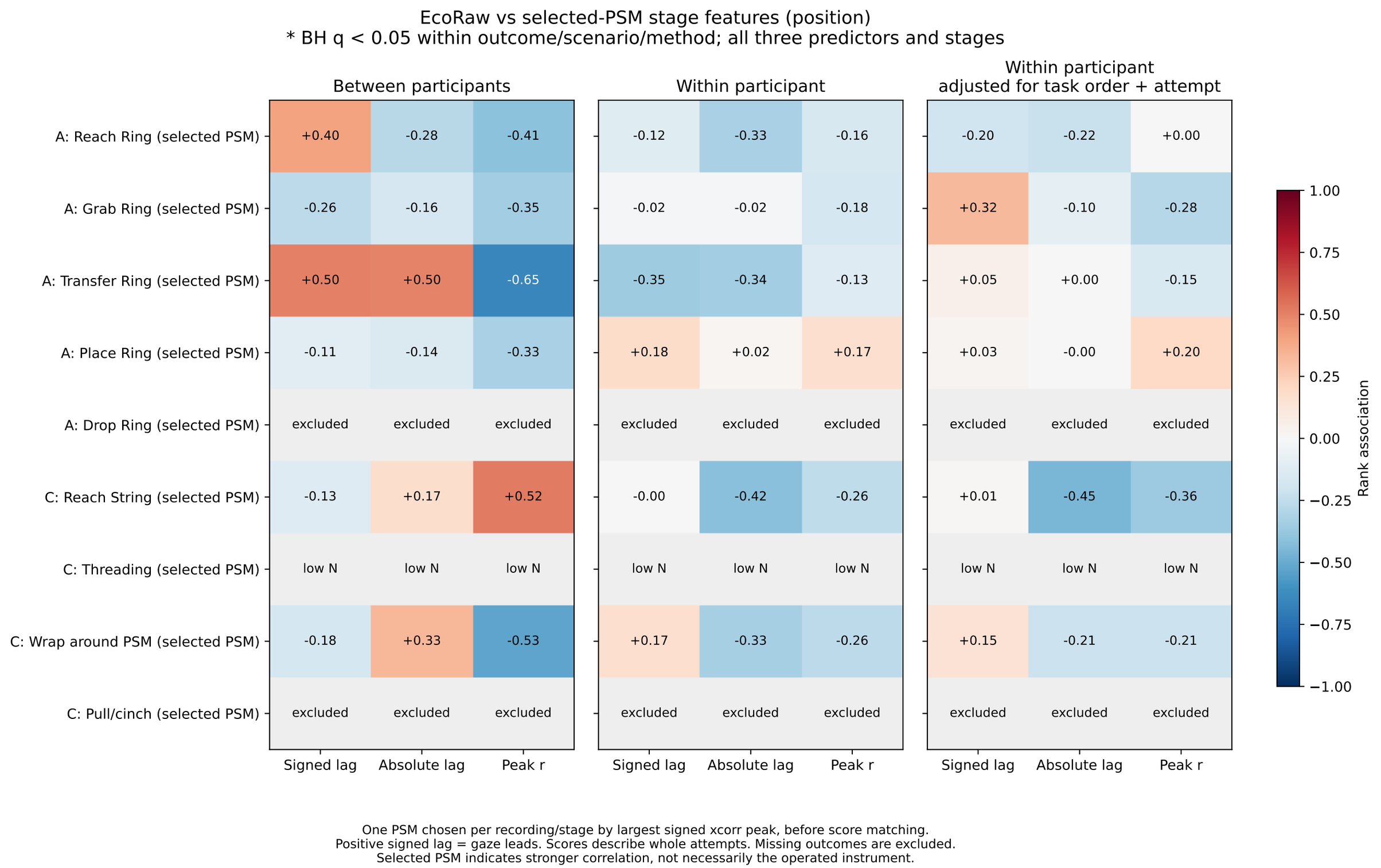


One PSM chosen per recording/stage by largest signed xcorr peak, before score matching.
Positive signed lag = gaze leads. Scores describe whole attempts. Missing outcomes are excluded.
Selected PSM indicates stronger correlation, not necessarily the operated instrument.

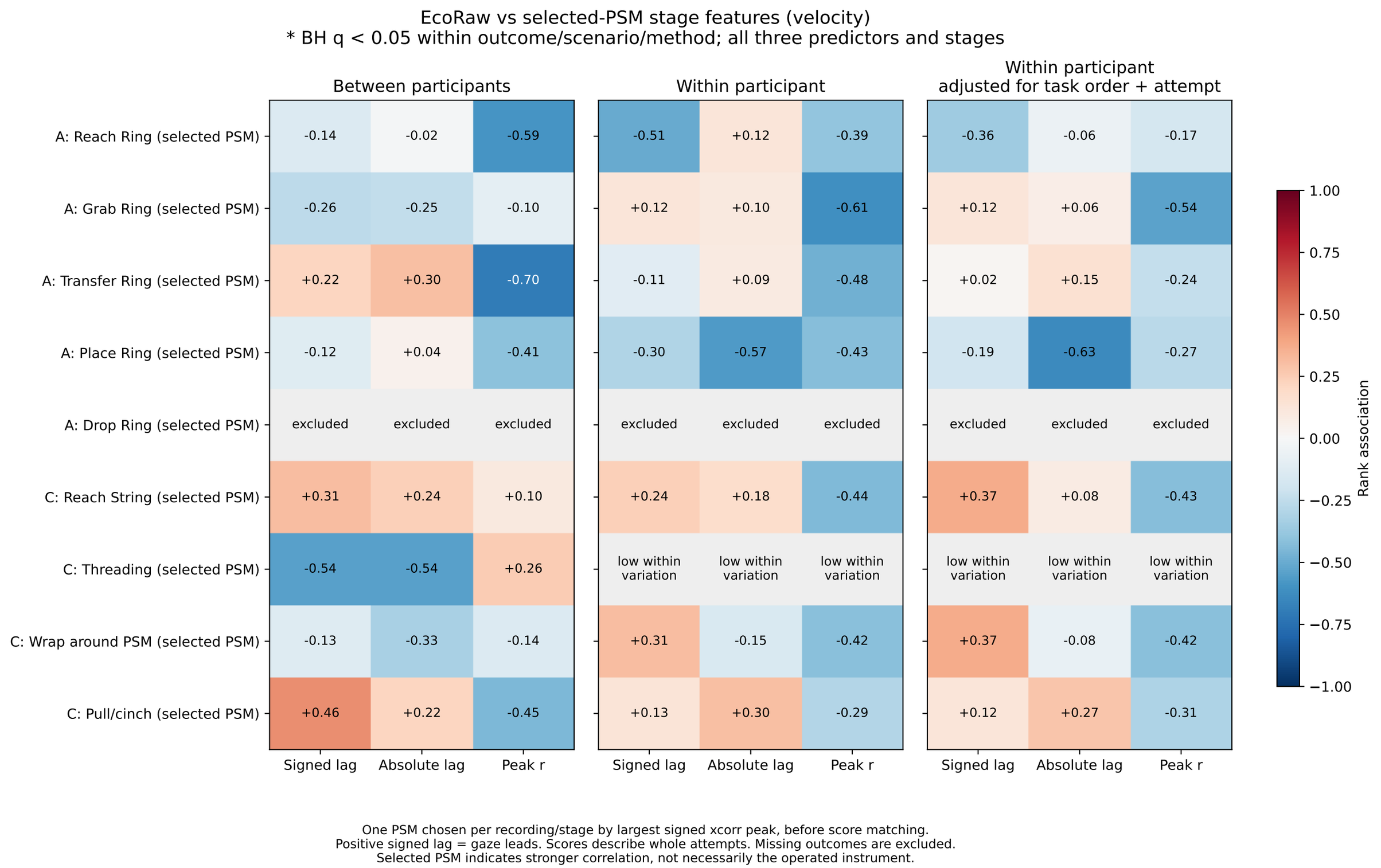

TimeRaw vs selected-PSM stage features (position)
* BH q < 0.05 within outcome/scenario/method; all three predictors and stages
Between participants
Within participant
Within participant
adjusted for task order + attempt
A: Reach Ring (selected PSM)
A: Grab Ring (selected PSM)
A: Transfer Ring (selected PSM)
A: Place Ring (selected PSM)
A: Drop Ring (selected PSM)
C: Reach String (selected PSM)
C: Threading (selected PSM)
C: Wrap around PSM (selected PSM)
C: Pull/cinch (selected PSM)
+0.29 -0.35 +0.33
-0.16 -0.27 -0.20
-0.29 -0.13 -0.04
+0.31 -0.18 -0.30
-0.37 -0.11 -0.16
-0.14 -0.27 -0.31
+0.73 +0.74 +0.08
-0.47 -0.37 +0.01
-0.11 -0.00 +0.07
+0.01 -0.02 +0.16
+0.16 +0.01 +0.12
-0.04 -0.02 +0.14
excluded excluded excluded
excluded excluded excluded
excluded excluded excluded
+0.14 +0.14 +0.38
+0.02 -0.35 -0.29
+0.03 -0.39 -0.39
low N low N low N
low N low N low N
low N low N low N
-0.27 +0.15 -0.77
-0.00 -0.35 -0.36
-0.04 -0.25 -0.32
excluded excluded excluded
excluded excluded excluded
excluded excluded excluded
Signed lag
Absolute lag
Peak r
Rank association
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
One PSM chosen per recording/stage by largest signed xcorr peak, before score matching.
Positive signed lag = gaze leads. Scores describe whole attempts. Missing outcomes are excluded.
Selected PSM indicates stronger correlation, not necessarily the operated instrument.



One PSM chosen per recording/stage by largest signed xcorr peak, before score matching.
Positive signed lag = gaze leads. Scores describe whole attempts. Missing outcomes are excluded.
Selected PSM indicates stronger correlation, not necessarily the operated instrument.



One PSM chosen per recording/stage by largest signed xcorr peak, before score matching.
Positive signed lag = gaze leads. Scores describe whole attempts. Missing outcomes are excluded.
Selected PSM indicates stronger correlation, not necessarily the operated instrument.


EcoRaw vs selected-PSM stage features (velocity)
* BH q < 0.05 within outcome/scenario/method; all three predictors and stages
Between participants
Within participant
Within participant
adjusted for task order + attempt
A: Reach Ring (selected PSM)
A: Grab Ring (selected PSM)
A: Transfer Ring (selected PSM)
A: Place Ring (selected PSM)
A: Drop Ring (selected PSM)
C: Reach String (selected PSM)
C: Threading (selected PSM)
C: Wrap around PSM (selected PSM)
C: Pull/cinch (selected PSM)
-0.14 -0.02 -0.59
-0.26 -0.25 -0.10
+0.22 +0.30 -0.70
-0.12 +0.04 -0.41
excluded excluded excluded
+0.31 +0.24 +0.10
-0.54 -0.54 +0.26
-0.13 -0.33 -0.14
+0.46 +0.22 -0.45
-0.51 +0.12 -0.39
+0.12 +0.10 -0.61
-0.11 +0.09 -0.48
-0.30 -0.57 -0.43
excluded excluded excluded
+0.24 +0.18 -0.44
low within variation
+0.31 -0.15 -0.42
+0.13 +0.30 -0.29
-0.36 -0.06 -0.17
+0.12 +0.06 -0.54
+0.02 +0.15 -0.24
-0.19 -0.63 -0.27
excluded excluded excluded
+0.37 +0.08 -0.43
low within variation
+0.37 -0.08 -0.42
+0.12 +0.27 -0.31
Signed lag
Absolute lag
Peak r
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Rank association
One PSM chosen per recording/stage by largest signed xcorr peak, before score matching.
Positive signed lag = gaze leads. Scores describe whole attempts. Missing outcomes are excluded.
Selected PSM indicates stronger correlation, not necessarily the operated instrument.